

Intrusion Detection on UNSW-NB15

Binary and Multiclass Classification of Network Traffic

Simone Montagnini

University of Bologna

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Project Objectives

This project addresses intrusion detection on network traffic through two classification tasks:

- **Binary classification:** distinction between normal and malicious traffic (`label`)
- **Multiclass classification:** identification of the specific threat among 9 possible types (`attack_cat`)

Source

Kaggle competition: *Advanced AI for Cybersecurity — Intrusion Detection with UNSW-NB15*

The Dataset — UNSW-NB15

- Realistic network traffic dataset with both normal and malicious network activity
- **175,341** training records, **42** original features (categorical and numerical)
- Two target columns:
 - label: 0 = normal, 1 = attack
 - attack_cat: Normal, Generic, Exploits, Fuzzers, DoS, Reconnaissance, Analysis, Backdoor, Shellcode, Worms

Target Distribution

Binary (label)

- Attack: 68.1%
- Normal: 31.9%

Multiclass (attack_cat)

- Normal: 56,000
- Generic: 40,000
- Exploits: 33,393
- Fuzzers: 18,184
- DoS: 12,264
- Reconnaissance: 10,491
- Analysis: 2,000
- Backdoor: 1,746
- Shellcode: 1,133
- **Worms: 130**

Balancing

The binary target is reasonably balanced, but `attack_cat` is severely imbalanced: Worms has $430\times$ fewer examples than Normal — need for **class-weighting** strategies in the multiclass model.

- **Duplicate rows:** 78,519 out of 175,341 (**44.8%**)
 - Removed before splitting, to avoid data leakage between train and test
- **Categorical features profile:**
 - proto (High cardinality): 133 unique values (tcp/udp $\approx$ 82% of traffic)
 - service: 13 values, 54% missing/unknown (-)
 - state: 9 values

Skewed Numerical Features

Several traffic features show extreme right-skewed distributions:

Feature	Skewness
trans_depth	167.3
response_body_len	76.3
sbytes	45.3
sloss	44.8
dloss	41.4
spkts	40.2

27 features exceeded the skewness threshold (> 1.0) and were logarithmic transformed ($\log_{10}$) in preprocessing.

Data Preparation & Transformation

- ① **Duplicate removal:** removed 78,519 duplicate rows before splitting
- ② **Train/test split:** 67/33, stratified on label
- ③ **Encoding:** OneHotEncoder on proto, service, state — 185 total features
- ④ **Skewness correction:** log1p transform on 27 features (skewness > 1.0)

Key methodological choice

Preprocessing steps are **fit exclusively on the training set** and mapped to the test set via `transform()` in order to prevent data leakage.

- **Strict separation:** Fitting encoders on the entire dataset injects future knowledge into the training phase
- **Handling unseen data:** Rare categories (e.g. within the 133 proto values) might appear exclusively in the test set
- **Fault tolerance:** `OneHotEncoder` is configured with `handle_unknown='ignore'` to safely bypass novel categories without causing execution failures

- **Supervised feature selection:** each feature is scored against the target, keeping only the top k features
- **Score function:** ANOVA F-value (`f_classif`)
 - `mutual_info_classif` was excluded due to extreme computation times on this dataset's dimensionality.
- **Selection:** $k = 50$ out of 185 features, selected separately for the `label` and `attack_cat` targets

Interpretability Preservation

SelectKBest retains the original traffic attributes. This ensures the model remains transparent, enabling direct attribution of malicious activity to specific network signals.

- **Prerequisite scaling:** features standardized (StandardScaler), then PCA fitted on the training set only
- **Explained variance:** retaining 90% of the variance requires **137 components** (118 for 80%)
- **Limited reduction:** 137 components out of 185 features

Component Analysis & Feature Contribution

Rather than being dominated by the numerous one-hot encoded columns proto, the principal components exhibit a mix of underlying data types. For instance, PC2 is driven almost entirely by timing and byte-related numerical features (`sinpkt`, `dload`, `tcprtt`)

Feature Selection Comparison

	SelectKBest	PCA
Dimensionality	50 features	137 components
Interpretability	High (original domain)	Low (linear combinations)
Transformation	Supervised (target-aware)	Unsupervised
Training efficiency	Acceptable	Impractical

Design Choice

PCA was excluded from the classification pipeline: even with a reduced number of components, the computational overhead during GridSearchCV was impractical. Classification proceeds with SelectKBest only.

Classification Methodology

- **Algorithms:** Decision Tree (as a baseline) and Random Forest
- **Hyperparameter Tuning:** `max_depth` and `criterion` for Decision Tree, `n_estimators` and `max_depth` for Random Forest, `class_weight` `None` vs `'balanced'` for both
- **Imbalance Handling:** Included `class_weight` (`None` vs `'balanced'`) to evaluate the best strategy for rare threat vectors
- **Validation:** 5-fold StratifiedKFold to ensure proportional representation of all classes during training

Evaluation Metrics

Accuracy is misleading on highly imbalanced datasets. Models are evaluated using **F1-Score** (Binary classification) and **Macro F1-Score** (Multiclass classification) to weigh the performance on minority classes equally.

Binary Classification Results

Model Performance

Metric	DT	RF
Accuracy	0.916	0.920
Precision	0.918	0.923
Recall	0.916	0.920
F1-Score	0.916	0.920

DT = Decision Tree, RF = Random Forest

- Random Forest achieves an excellent F1-Score of **0.920**.
- The model is **highly reliable** in distinguishing malicious activity from normal traffic.

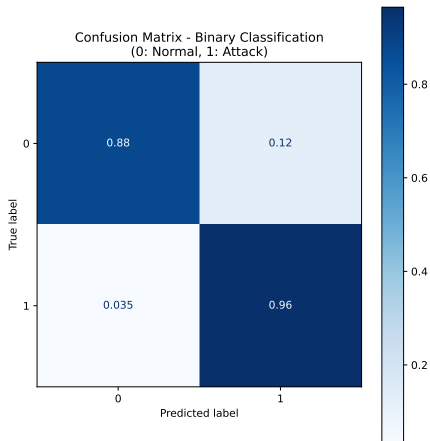


Figure: Binary Confusion Matrix

Multiclass Classification Results

The Multiclass Challenge

- Best model achieves a **Macro F1-Score** of **0.57**
- **Majority classes** (Normal, Generic) are predicted with high confidence
- **Minority classes** drag the macro average down due to overlapping feature spaces and extreme sparsity

Impact of Balancing

Thanks to `class_weight='balanced'`, the model is also able to detect instances of ultra-rare attacks (like Worms), preventing them from being completely ignored.

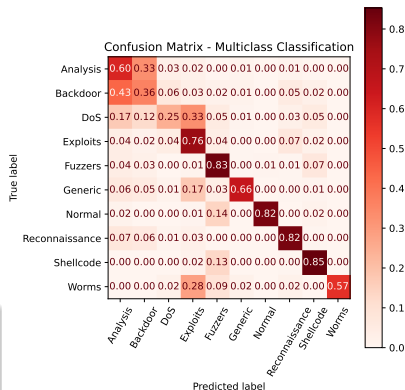
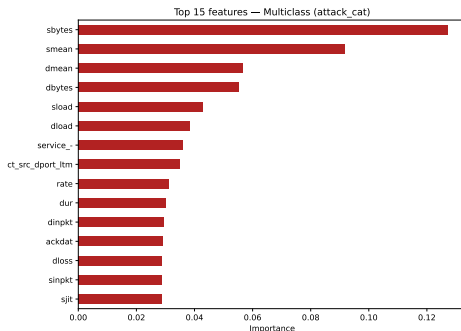
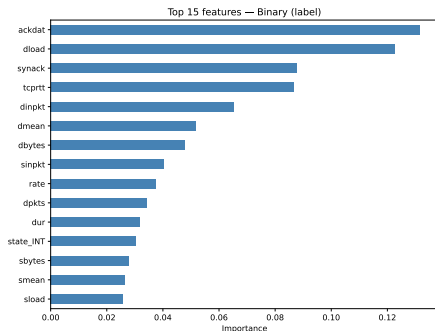


Figure: Multiclass Confusion Matrix

Feature Importance Analysis

- **Binary task:** Mostly relies on TCP handshake timing features (ackdat, synack, tcprtt), anomalous handshakes are a strong indicator of malicious traffic
- **Multiclass task:** Based on byte-transfer volume features (sbytes, smean, dbytes) to differentiate the *type* of attack



- **Methodological trade-offs:** Prioritizing interpretability (SelectKBest and Tree-based models) proved highly effective for binary detection, achieving an F1-Score of 0.92 while maintaining transparent decision rules
- **The Multiclass bottleneck:** Differentiating specific attack vectors remains intrinsically difficult (Macro F1: 0.57). The confusion matrix revealed concentrated overlap between related categories (e.g. DoS/Exploits, Backdoor/Analysis) and, notably, between Fuzzers and Normal traffic, independent of class imbalance
- **Data Quality Dependency:** The project highlighted how extreme class imbalance heavily constrain the theoretical upper bound of any standard classifier's performance

To overcome the current limitations and push the system toward a realistic Network Intrusion Detection System, future iterations could explore:

- **Advanced Resampling:** Although SMOTE was initially considered, it was discarded due to extreme computational overhead. Future work must investigate highly optimized, scalable sampling techniques
- **Alternative Architectures:** Evaluating non-tree-based classifiers to determine if different mathematical boundaries can better isolate rare classes
- **Domain-Specific Feature Engineering:** Deriving new, highly specific network features aimed at better distinguishing the overlapping signatures of minority attacks from more frequent anomalies